

THE INDEPENDENT INSTITUTE OF EDUCATION

RaceDay

API Endpoint Plan

Portfolio of Evidence — Part 1: System Planning and Database

MODULE NAME: PROGRAMMING 2B

MODULE CODE: PROG6212/w

STUDENT NAME:

STUDENT NUMBER:

RaceDay API Endpoint Plan

This plan is built directly from the RaceDay ERD and `raceday_schema.sql`, and covers every functional requirement described in Part 2 of the POE: Authentication, User Profile, Events, Categories, Event Enrolments, and Results.

Role legend **None** = public / no login required · **Any** = any authenticated user · **Organiser** / **Participant** = role-restricted.

Where an endpoint is marked “(owner)”, the API must also confirm the logged-in Organiser or Participant owns the specific record being acted on — not just that they hold the required role.

1. Authentication

Public endpoints that allow a new user to register as either an Organiser or a Participant, and for a registered user to start or end a session.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	<code>/api/auth/register</code>	Creates a new account as either an Organiser or a Participant.	None	{ firstName, lastName, email, password, phoneNumber, role }	201 Created – new user (no password returned) 400 Bad Request – validation failed 409 Conflict – email already registered
POST	<code>/api/auth/login</code>	Verifies credentials and starts a session that stores the user's ID and role.	None	{ email, password }	200 OK – user summary + role, session cookie set 401 Unauthorized – invalid email or password
POST	<code>/api/auth/logout</code>	Ends the current session.	Any	None	200 OK – logged out 401 Unauthorized – no active session

2. User Profile

Both roles can view and update their own profile information.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	<code>/api/users/me</code>	Returns the logged-in user's own profile.	Any	None	200 OK – profile object 401 Unauthorized

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
PUT	/api/users/me	Updates the logged-in user's own profile.	Any	{ firstName, lastName, phoneNumber, profilePictureUrl }	200 OK – updated profile 400 Bad Request – validation failed 401 Unauthorized

3. Events

Organisers manage events end-to-end; both roles can browse and view event details.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Lists all events, with optional query filters (type, location, fromDate).	None	None	200 OK – array of events
GET	/api/events/{id}	Returns full details of one event, including its categories and routes.	None	None	200 OK – event detail object 404 Not Found
POST	/api/events	Creates a new event owned by the logged-in Organiser.	Organiser	{ name, description, eventDate, location, distanceKm, eventType, bannerImageUrl }	201 Created – new event 400 Bad Request 403 Forbidden – not an Organiser
PUT	/api/events/{id}	Updates an event (owner Organiser only).	Organiser (owner)	{ name, description, eventDate, location, distanceKm, eventType, bannerImageUrl }	200 OK – updated event 400 Bad Request 403 Forbidden – not the owning Organiser 404 Not Found
DELETE	/api/events/{id}	Deletes an event (owner Organiser only).	Organiser (owner)	None	204 No Content 403 Forbidden 404 Not Found

4. Categories

Age or distance categories are defined per event by its Organiser; both roles can view them.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Lists the age/distance categories available for an event.	None	None	200 OK – array of categories 404 Not Found – event does not exist
POST	/api/events/{eventId}/categories	Adds a category to an event.	Organiser (owner)	{ categoryName, minAge, maxAge, distanceKm }	201 Created – new category 400 Bad Request 403 Forbidden 404 Not Found
PUT	/api/categories/{id}	Updates a category's details.	Organiser (owner)	{ categoryName, minAge, maxAge, distanceKm }	200 OK – updated category 400 Bad Request 403 Forbidden 404 Not Found
DELETE	/api/categories/{id}	Removes a category from an event.	Organiser (owner)	None	204 No Content 403 Forbidden 404 Not Found 409 Conflict – active enrolments exist

5. Event Enrolments

Participants enrol into an event under a chosen category; Organisers view enrolments for events they own.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{eventId}/enrolments	Enrols the logged-in Participant into an event under a chosen category.	Participant	{ categoryId }	201 Created – new enrolment 400 Bad Request 401 Unauthorized 404 Not Found – event/category missing 409 Conflict – already enrolled
GET	/api/users/me/enrolments	Returns all events the logged-in Participant has enrolled in, with status.	Participant	None	200 OK – array of enrolments 401 Unauthorized

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/enrolments	Returns all Participants enrolled in an event the Organiser owns.	Organiser (owner)	None	200 OK – array of enrolments 403 Forbidden 404 Not Found
DELETE	/api/enrolments/{id}	Cancels the logged-in Participant's own enrolment, provided the event has not started.	Participant (owner)	None	204 No Content 403 Forbidden – not the owning Participant 404 Not Found 409 Conflict – event already started

6. Results

Organisers capture finish times and positions after an event; Participants view their own results.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{eventId}/results	Captures/publishes a finish time and position for an enrolled Participant after the event.	Organiser (owner)	{ enrolmentId, finishTimeSeconds, finishPosition, totalFinishers }	201 Created – new result 400 Bad Request 403 Forbidden 404 Not Found – event/enrolment missing
PUT	/api/results/{id}	Corrects a previously captured result.	Organiser (owner)	{ finishTimeSeconds, finishPosition, totalFinishers }	200 OK – updated result 400 Bad Request 403 Forbidden 404 Not Found
GET	/api/users/me/results	Returns the logged-in Participant's full personal race history.	Participant	None	200 OK – array of results with event name, date, category, finish time and position 401 Unauthorized
GET	/api/results/{id}	Returns one result (viewable by the owning Participant or the Organiser of that event).	Any (owner-checked)	None	200 OK – result object 403 Forbidden 404 Not Found

Coverage Check Against Part 2 Functional Requirements

Authentication: register + login

User Profile: view + update own profile, both roles

Events: full CRUD for Organisers, read access for both roles

Categories: full CRUD for Organisers, read access for both roles

Event Enrolments: Participant enrolment, Organiser visibility of enrolments

Results: Organiser capture, Participant personal view